

Notes: Ellison & Ellison (Econometrica, 2009)

Search, Obfuscation, and Price Elasticities on the Internet

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Pricewatch market environment	4
3.2	Price and rank data	4
3.3	Sales, price, and cost data from the retailer	4
3.4	Main variables	5
4	Models and empirical specifications	5
4.1	Theory block 1: incomplete consumer search	5
4.2	Theory block 2: add-ons, upgrades, and adverse selection	6
4.3	Demand estimation	6
4.4	Elasticity decomposition	6
4.5	Conditional site-choice logit	7
4.6	Instrumental-variable estimates	7
4.7	Markup logic	7
5	Identification and empirical design	7
5.1	What identifies the demand effects	7
5.2	Why rank is especially useful	8
5.3	Main threats and responses	8
6	Section-by-section study guide	8
7	Tables and figures	9
7.1	Figure 1: A Sample Pricewatch Search List	9
7.2	Figure 2: A Website Designed to Induce Upgrading	9
7.3	Figure 3: Another Website Designed to Induce Upgrades and Add-ons	10
7.4	Table I: Summary Statistics for Memory Module Data	11
7.5	Table II: Demand for 128MB PC100 Memory Modules	12
7.6	Table III: Price Elasticities for Memory Modules	12
7.7	Table IV: Evidence of Incomplete Consumer Learning	13
7.8	Table V: Instrumental Variables Estimates of PC100 128MB Memory Demand	13
7.9	Table VI: Mean Percentage Markup in Six Product Classes	14

8	Result map	14
9	Short summary	15
10	Conclusion	15
10.1	Core conclusion	15
10.2	Economic intuition	15
10.3	Why the evidence is persuasive	16
10.4	Incremental contribution to the literature	16
10.5	Limitations	16
10.6	Takeaway	16

1 Big picture

- **Question.** Does Internet price search push markets toward frictionless competition, or do firms respond by making search less effective?
- **Setting.** The paper studies online computer-parts retailers listed on Pricewatch.com, a price-search engine that sorts sellers by posted item price. The empirical environment is unusually useful because the product categories are narrow, price lists are observed at high frequency, and the authors obtain actual sales and cost data from an online retailer.
- **Core friction.** Price search makes it easy to compare the advertised low price, but it does not fully reveal product quality, warranty terms, shipping terms, return policies, compatibility, or the full cost of upgrades and add-ons.
- **Obfuscation.** Retailers respond to search by designing product listings, product quality choices, and checkout pages that make the advertised low price less informative. The paper documents low-quality loss leaders, complicated product descriptions, shipping-cost games, return-policy differences, preselected add-ons, and upgrade menus.
- **First main empirical result.** Demand for low-quality products at the top of the Pricewatch list is extremely price sensitive. For 128MB PC100 low-quality memory, the estimated own-price elasticity is about -24.9 .
- **Second main empirical result.** Low posted prices for low-quality products help sell medium- and high-quality products. This is evidence of a loss-leader/upgrading mechanism: many consumers first use Pricewatch to locate a low-priced site, then switch within the site to a better product.
- **Incomplete learning result.** Even after a consumer has chosen to buy a medium- or high-quality module, the low-quality Pricewatch rank still predicts which of the retailer's two sites the consumer chooses. This indicates that consumers do not fully learn all relevant prices and terms.
- **Adverse-selection result.** The elasticity of demand for low-quality products is larger in absolute value than the elasticities for medium- and high-quality products. Lowering the advertised low-quality price attracts more low-margin shoppers and changes the quality mix, weakening incentives to cut the advertised price.
- **Markup result.** Naive inverse-elasticity logic would predict markups around 3–6 percent, but observed revenue-weighted markups are much larger, with overall category markups around 8–16 percent. Obfuscation helps reconcile high elasticities with positive markups.
- **Economic takeaway.** Search technology and obfuscation are strategic complements. Search engines make one price dimension transparent, but this induces firms to compete on the transparent dimension while shifting margins to opaque dimensions.

2 Incremental contribution of the paper

- **From price dispersion to actual demand.** Earlier Internet-pricing studies often used posted prices or clicks. This paper combines high-frequency search-engine listings with actual transaction quantities, prices, and costs. That lets the authors estimate demand elasticities and markups rather than only documenting price dispersion.
- **Empirical measurement of obfuscation.** The paper transforms obfuscation from a mostly theoretical idea into a measurable set of practices: low-quality advertised products, upgrades, add-ons, contractual complexity, shipping frictions, and quality opacity.
- **Evidence on loss leaders in a real online market.** The paper provides unusually clean evidence that the low advertised price of an inferior product increases sales of higher-quality products on the same site. This is a different mechanism from the classic supermarket loss leader because the goal is not necessarily to sell a second physical good, but to attract the consumer and then induce upgrading.

- **Link between search theory and add-on pricing.** The paper connects two mechanisms that had often been treated separately: incomplete consumer search and add-on/adverse-selection pricing. It then tests implications of both using the same micro data.
- **Demonstration of endogenous product degradation.** The paper shows that firms may intentionally create or emphasize inferior versions of products because search engines reward low advertised prices. The quality degradation is endogenous to the search technology.
- **Markup evidence.** By using cost data, the paper can ask whether the observed obfuscation is economically important enough to sustain markups. The answer is yes: observed markups are far above what a simple single-good elasticity rule would predict.
- **Broader contribution.** The paper reframes the Internet-commerce debate. It shows that the Internet does not mechanically eliminate market power; instead, firms redesign strategies around the search interface.

3 Data and measurement

3.1 Pricewatch market environment

- Pricewatch was a search engine for computer-parts shoppers. It returned lists of sellers sorted by posted item price.
- The lists display price, shipping charge, dealer name and phone number, state, and product/part identifiers. But they cannot fully summarize quality, compatibility, warranty, return policy, restocking fees, shipping details, or add-on menus.
- The authors study four memory-module categories: 128MB PC100, 128MB PC133, 256MB PC100, and 256MB PC133.
- PC100 versus PC133 refers to the memory speed; 128MB versus 256MB refers to capacity. The categories are not perfect substitutes, which helps keep the empirical product space manageable.

Interpretation: The search engine makes the advertised low-quality item price highly salient, but it cannot make the full product bundle transparent. This creates the room for obfuscation.

3.2 Price and rank data

- The authors repeatedly queried Pricewatch.com from May 2000 to May 2001.
- They collected the 12 or 24 lowest price offerings within each memory category at hourly frequency.
- Daily variables are constructed by averaging across hours using weights that reflect average hourly sales volume.
- The key search-ranking variable is $\log(1 + PLowRank)$, where *PLowRank* is the website's position in the Pricewatch price-ranked list.

Interpretation: Rank is economically important because Pricewatch consumers are mechanically directed toward the top of the list. Small posted price changes can move a firm sharply up or down the list.

3.3 Sales, price, and cost data from the retailer

- The authors obtained transaction data from one Internet retailer that operates two websites, called Site A and Site B.

- Each site sells three quality tiers within each memory category: low, medium, and high.
- The websites have identical product lineups, which is important for conditional site-choice tests.
- The data are aggregated to website-day-product observations. In the 128MB PC100 category, there are 683 website-day observations.
- The firm also supplied wholesale acquisition costs, which are used for markup calculations and cost-based instruments.

Interpretation: The combination of search-engine list positions, actual sales quantities, and cost data is what allows the paper to move from qualitative examples to demand and markup evidence.

3.4 Main variables

- P_{Low} : transaction price for the low-quality memory module on a site-day.
- P_{Mid} : transaction price for the medium-quality module.
- P_{Hi} : transaction price for the high-quality module.
- *LowestPrice*: lowest price appearing on the Pricewatch list in the category.
- $P_{LowRank}$: daily average Pricewatch rank of the site.
- Q_{Low}, Q_{Mid}, Q_{Hi} : daily quantities sold by quality tier.
- *SiteB*: indicator for one of the two websites.
- Weekend and piecewise linear time trends control for demand seasonality and smooth shifts.

Interpretation: The analysis distinguishes three effects: the site’s own posted prices, the search-engine list rank generated by low posted prices, and broader category-level price trends.

4 Models and empirical specifications

4.1 Theory block 1: incomplete consumer search

The first theoretical channel is standard search. Consumers may not inspect all relevant prices and terms before buying. Obfuscation increases search costs or the fraction of consumers who face search costs.

- Fully informed consumers buy from the cheapest relevant seller.
- Partially informed consumers may stop searching too early and buy from a higher-priced or more opaque seller.
- A retailer benefits from making consumer learning less complete, because incomplete learning reduces the discipline of price comparison.

Interpretation: This channel predicts that even consumers buying medium- or high-quality products may respond to the low-quality Pricewatch rank. If rank matters after conditioning on the actual product price, consumers are not fully learning all prices and terms.

4.2 Theory block 2: add-ons, upgrades, and adverse selection

Each firm sells a low-quality good L and a high-quality good H . Firms post low-quality prices on the search engine and choose high-quality/upgrade prices on their websites. Let p_{iL} be firm i 's posted low-quality price and p_{iH} its high-quality price. Let the incremental upgrade price be

$$p_i^m = p_{iH} - p_{iL},$$

and the incremental cost be

$$c^u = c_H - c_L.$$

Let $x(p, p_{iL}, p_{-iL})$ be the fraction of visitors who upgrade at upgrade price p . The upgrade price is set at the ex post monopoly level:

$$p_i^{m*}(p_{iL}, p_{-iL}) = \arg \max_p (p - c^u) x(p, p_{iL}, p_{-iL}).$$

Define $x^*(p_{iL}, p_{-iL})$ as the equilibrium fraction of visitors who upgrade. Firm i 's profit from a low-quality posted price is:

$$\Pi_i(p_{iL}, p_{-iL}^*) = [p_{iL} - c_L + x^*(p_{iL}, p_{-iL}^*) (p_i^{m*}(p_{iL}, p_{-iL}^*) - c^u)] D_i(p_{iL}, p_{-iL}^*).$$

Interpretation: The low posted price affects profit through two margins. It attracts more visitors, but it also changes the mix of visitors. If a lower low-quality price attracts more low-quality buyers who are less likely to upgrade, then price cuts become less attractive than a single-good demand elasticity would imply.

4.3 Demand estimation

For category c , website w , quality q , and day t , the authors estimate a multiplicative demand model:

$$Q_{wcqt} = \exp(X_{wct}\beta_{cq})u_{wcqt},$$

where

$$\begin{aligned} X_{wct}\beta_{cq} = & \beta_{cq0} + \beta_{cq1} \log(P_{Lowwct}) + \beta_{cq2} \log(P_{Midwct}) + \beta_{cq3} \log(P_{Hiwct}) \\ & + \beta_{cq4} \log(LowestPrice_{ct}) + \beta_{cq5} \log(1 + P_{LowRank}_{wct}) + \beta_{cq6} Weekend_t + \beta_{cq7} SiteB_w + \sum_s \delta_{cqs} Trend_{st}. \end{aligned}$$

The GMM moment condition is:

$$E [Q_{wcqt} e^{-X_{wct}\beta_{cq}} - 1 \mid X_{wct}] = 0.$$

Interpretation: This is a semi-log quantity equation with multiplicative errors. It estimates own-price, cross-price, and rank effects while controlling for time trends and site effects.

4.4 Elasticity decomposition

For low-quality products, the own-price elasticity includes two effects:

- a direct effect of $\log(P_{Low})$ on demand;
- an indirect effect through $\log(1 + P_{LowRank})$, because a lower low-quality price improves rank.

Interpretation: This decomposition matters because the extreme price sensitivity of low-quality products comes mostly from the rank channel. Consumers do not necessarily react to a few dollars mechanically; they react to the position created by those dollars on the search-engine list.

4.5 Conditional site-choice logit

For buyers of medium- and high-quality products, the paper estimates logit regressions in which the dependent variable is an indicator for choosing Site A rather than Site B. Explanatory variables are differences across the two sites in:

$$\Delta \log(1 + P_{LowRank}), \quad \Delta \log(P_{Mid}), \quad \Delta \log(P_{Hi}),$$

plus controls.

Interpretation: Since Sites A and B sell identical products, a fully informed medium-quality buyer should mainly choose based on the medium-quality price. If the low-quality rank still predicts site choice, then rank is guiding consumers even when the purchased product is not the low-quality item.

4.6 Instrumental-variable estimates

The authors use two instrument sets to address price endogeneity:

- **Cost instruments:** wholesale acquisition costs for low-, medium-, and high-quality products.
- **Other-speed instruments:** prices and ranks in the related memory-speed category, such as PC133 prices used as instruments for PC100 prices.

Interpretation: The IV estimates are noisier, but the main qualitative finding remains: low-quality rank has a strong effect on demand. The price variables are harder to identify precisely because instruments are imperfect and the sample is small.

4.7 Markup logic

Absent adverse selection, a simple inverse elasticity rule would imply approximate markups:

$$Markup \approx \frac{1}{|\varepsilon|}.$$

The paper then compares this to observed revenue-weighted markups and to predicted markups incorporating the adverse-selection multiplier.

Interpretation: With elastic demand, naive markups should be small. Higher observed markups indicate that obfuscation weakens the connection between the advertised low price and total profitability.

5 Identification and empirical design

5.1 What identifies the demand effects

- The high-frequency Pricewatch data generate frequent changes in list position.
- The retailer's two sites change posted prices discontinuously while demand trends and rival prices move more smoothly.
- Piecewise time trends absorb common market movements in memory demand and wholesale price cycles.
- Website fixed effects control for persistent differences across Site A and Site B.

Interpretation: Identification comes from within-product, within-website time-series variation in prices and ranks, not from cross-sectional differences across retailers.

5.2 Why rank is especially useful

- Pricewatch rank is mechanically generated by posted low-quality prices and rivals' posted prices.
- Retailers manually update prices, so ranks drift and jump as firms respond at different times.
- The authors argue that idiosyncratic short-run rank variation is plausibly not targeted to daily demand shocks.

Interpretation: The rank variable captures the search-engine mechanism directly: moving down the list dramatically reduces visibility and therefore sales.

5.3 Main threats and responses

- **Price endogeneity.** Demand shocks may raise prices and quantities simultaneously, biasing own-price elasticities toward zero. The authors use flexible trends and IV estimates.
- **Rival prices.** The retailer's prices may be correlated with rival prices. LowestPrice and rank controls help, and IV estimates provide a robustness check.
- **Single-firm data.** Quantities and costs come from one retailer. The setting is rich but not representative of all Internet retail markets.
- **Measurement of quality.** Quality tiers bundle physical product quality and contractual terms. This is exactly the obfuscation margin, but it complicates clean product comparisons.

Interpretation: The paper is strongest as micro evidence on a concrete search-engine market. Its purpose is not to estimate universal online retail elasticities but to show the mechanisms by which search and obfuscation interact.

6 Section-by-section study guide

- **Introduction.** Defines the puzzle: the Internet lowers search costs but also creates incentives for obfuscation. Reports the headline elasticity and markup facts.
- **Theory of search and obfuscation.** Provides two conceptual frameworks: incomplete search and add-on/adverse-selection pricing.
- **Pricewatch environment and obfuscation examples.** Shows actual screenshots and describes how retailers use low prices, quality complexity, shipping fees, and add-ons.
- **Data.** Explains how hourly Pricewatch data are matched to retailer sales and cost data.
- **Demand patterns.** Estimates elasticity matrices and documents that low-quality rank affects all quality tiers.
- **Incomplete learning tests.** Uses Site A versus Site B choices to test whether consumers fully learn medium/high product prices.
- **IV estimates.** Checks whether demand results survive alternative price instruments.
- **Markups.** Compares observed markups to single-good inverse-elasticity predictions and adverse-selection-adjusted predictions.
- **Conclusion.** Argues that firms endogenously design product quality and complexity in response to search-engine incentives.

7 Tables and figures

7.1 Figure 1: A Sample Pricewatch Search List

Read:

- The figure shows the first page of Pricewatch results for 128MB PC100 memory modules at a specific time.
- Sellers are ordered by posted item price, with columns for product description, price, shipping, dealer, date/time, and part number.
- Many listings are difficult to compare because descriptions and terms differ substantially.
- Even on a price-ranked list, the full price-quality bundle is opaque.

Detailed interpretation: The figure is the empirical starting point. It shows why the market is both highly transparent and highly opaque. The price dimension is transparent because products are sorted mechanically by item price. But the product descriptions, warranties, shipping terms, restocking conditions, and part numbers are heterogeneous. A consumer looking only at the list can identify a cheap offer, but cannot easily identify the best total deal.

Economic message: Price-search technology makes one attribute prominent. Firms then have incentives to shift competition toward less visible attributes.

BRAND	PRODUCT	DESCRIPTION	PRICE	SHIP	DATE/TIME	DEALER/PHONE	ST	PARTS
Generic	PRICE FOR ONLINE ORDERS ONLY - 128MB PC100 SDRAM DIMM - Bns Gold leads	* LIMIT ONE Easy installation - in stock	\$ 68	9.99 INSURED	10/12/00 12:40:05 AM CST	Computer.CiaB Inc. 800-487-4910 727-327-7559 Online Ordering	FL	MEM-128-100PCT
Generic	ONLINE ORDERS ONLY - 128MB SDRAM PC100 16x64 168pin	* LIMIT ONE	\$ 69	INSURED\$9.95	10/11/00 10:59:56 PM CST	Connect Computers 888-277-6287 949-367-0703 Online Ordering	CA	-
Generic	PRICE FOR ONLINE ORDER - 128MB PC100 SDRAM DIMM	* LIMIT ONE - inStock, 16x64 Gold Leads	\$ 70	10.75	10/11/00 2:11:00 PM CST	1st Choice Memory 949-888-3810 - P.O. % accepted Online Ordering	CA	-
Generic	PRICE FOR ONLINE ORDER - 128MB True PC100 SDRAM EPPROM DIMM 16x64 168pin Bns Gold Leads	* LIMIT ONE - in stock - with Lifetime Warranty	\$ 72	9.85	10/10/00 11:30:39 AM CST	ackson.com 800-382-6678 - P.O. % accepted Online Ordering	CA	-
Generic	IN STOCK, 128MB PC100 3.3vlt unbuffered SDRAM Gold Lead 168 Pin, 7/8ins - with Lifetime warranty	* LIMIT ONE Not compatible with E Machine	\$ 74	10.95- UPS INSURED	10/11/00 12:44:00 PM CST	Memulus.com 877-918-6767 626-918-6767	CA	880060
Generic	PRICE FOR ONLINE ORDERS ONLY - 128MB True PC100 SDRAM DIMM - Bns Gold - warranty	* LIMIT ONE	\$ 74	10.25	10/9/00 6:53:25 PM CST	FastRAM 800-487-1327	CA	-
House Brand	128MB PC100 3.3vlt SDRAM 168 Pin, 7/8ins - with LIFETIME WARRANTY	* LIMIT ONE	\$ 74	10.50 FedEx	10/11/00 10:20:23 AM CST	1st Comm Choice 800-345-8880 800-345-8880	OH	-
Generic	128MB 168Pin TRUE PC100 SDRAM - OEM 168pin	DIMM16x64 168pin Bns/7ns Gold Leads	\$ 75	\$10	10/11/00 2:37:00 PM CST	Sunset Marketing, Inc. 800-397-5050 410-628-0111 - P.O. % accepted Online Ordering	MD	-
Generic	128MB 16x64 PC100 Bns SDRAM.	* LIMIT ONE	\$ 77	10.90	10/12/00 9:37:45 AM CST	PC COST 800-877-9442 847-690-0103 Online Ordering	IL	-
Generic	IN STOCK, PC100, 128MB, 168pins DIMM NonECC, - with Lifetime warranty	* LIMIT 5	\$ 77	\$10.95 & UP For UPS Ground	10/9/00 5:11:10 PM CST	Augusta Technology, Inc 877-468-5181 909-468-1883 Online Ordering	CA	-
Generic	128MB PC100 BNS 16x64 SDRAM - one year warranty	* LIMIT ONE	\$ 78	UPS Ground \$10.62	10/11/00 2:16:36 PM CST	Computer Source 800-305-4930 847-640-9710 Online Ordering	IL	-
Generic	PRICE FOR ONLINE ORDERS ONLY - PC100 128MB NonBuffered, NonECC 16x64 SDRAM DIMM 3.3V Bns mem module	* LIMIT ONE - with lifetime warranty	\$ 78	10.95	10/5/00 6:29:59 PM CST	Jazz Technology USA, LLC 888-483-8872 909-869-8859	CA	ME-GBP100128

Page 1 of 10
1 5 10 15 20 25 30 Next

FIGURE 1.—A sample Pricewatch search list: 128MB PC100 memory modules at 12:01pm ET on October 12, 2000.

Figure 1: A sample Pricewatch search list.

7.2 Figure 2: A Website Designed to Induce Upgrading

Read:

- The figure shows three versions of a memory module with side-by-side features.

Memory Spec. Chart - PC3200 DDR 512MB (Select Your Memory Module)		
☐ Samsung/Micron or Major 512MB PC3200 [ADD \$25]	☐ Industry Standard 512MB PC3200 [ADD \$15]	☒ OEM 512MB PC3200
- CAS 2.5 Latency - Hand Picked 5ns 6 Layer Low Noise Shielded PCB Board 32x8 DRAM Type - Samsung/Micron or Major Brands - Return Shipping Paid - No Restocking Fee - Satisfaction & Compatibility Guaranteed - Lifetime Warranty 15 Days Full Refund - Memory Tested Before Ship Out - Copper Heat Sink - Cool Down the Memory up to 40%	- CAS 2.5 Latency - Hand Picked 5ns 6 Layer Low Noise Shielded PCB Board 32x8 DRAM Type - Industry Standard DRAM Chips 7 Days No Restocking Fee - Return Shipping not Paid - Improved Compatibility - Lifetime Warranty - Aluminum Heat Sink - Cool Down the Memory up to 35%	- CAS 3 Latency 4 Layer Module Board 64x4 DRAM Type - OEM DRAM Downgrade Chips 20% Restocking Fee According to the Market Value - Verify Compatibility with Memory Configurator - Return Shipping not Paid 9 Months Warranty

FIGURE 2.—A website designed to induce consumers to upgrade to a higher quality memory

Figure 2: A website designed to induce consumers to upgrade to a higher-quality memory module.

- The lowest-price product is shown with inferior characteristics and worse contract terms.
- Consumers are nudged toward higher-quality options with add-on prices such as “ADD \$15” or “ADD \$25.”
- Comparisons within the site are easier than comparisons across sites.

Detailed interpretation: This figure illustrates the loss-leader mechanism. The retailer can appear high on Pricewatch by advertising a low-quality product. Once the consumer arrives, the retailer presents a menu in which better quality, better warranty, lower restocking fees, and better compatibility all point toward upgrading. The retailer therefore competes aggressively for search-engine traffic but monetizes consumers after they arrive.

Economic message: Search engines can transform competition into a funnel: low advertised prices attract consumers, while opaque quality dimensions create room for upgrading and markups.

7.3 Figure 3: Another Website Designed to Induce Upgrades and Add-ons

Read:

- The figure shows checkout add-ons and upgrades that increase the selected price from \$53.81 to \$90.36.
- Some boxes are preselected, so avoiding add-ons requires active consumer effort.
- Add-ons include memory upgrades, cables, tools, cleaning kits, fans, pretesting, and cooling kits.
- The continuation page shows additional pretest and performance-enhancement options.

Detailed interpretation: Figure 3 makes the search-cost and add-on channels visually concrete. Even after a consumer has clicked through from the search list, the purchase path is complex. The consumer must understand multiple options and undo preselected choices. This makes the initially advertised price a weak statistic for the actual transaction price and gives the retailer a way to raise revenue without losing the search-engine ranking.

Economic message: Obfuscation does not require hiding the listed price. It can work by making the conversion from listed price to final purchase bundle cognitively and procedurally costly.

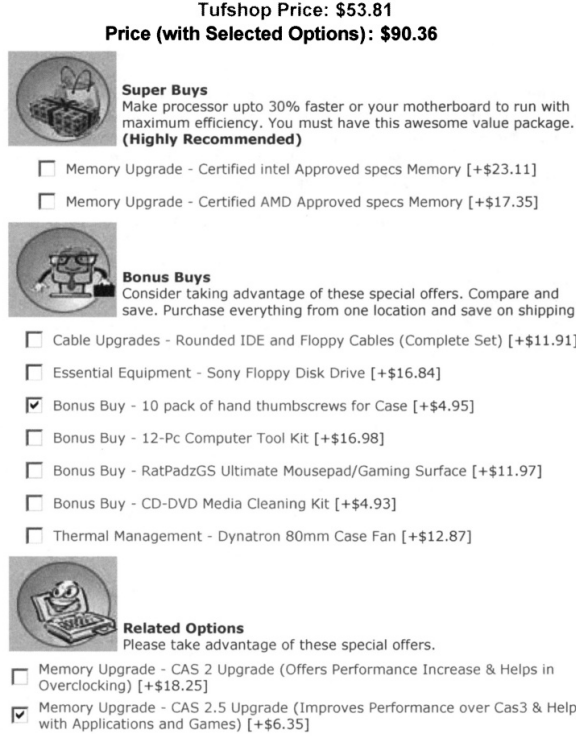


FIGURE 3.—Another website designed to induce consumers to upgrade and/or buy add-ons.

upgrades. To avoid purchasing the various add-ons, the consumer must read through the various options and unclick several boxes. After completing this

(a) Figure 3, part 1.

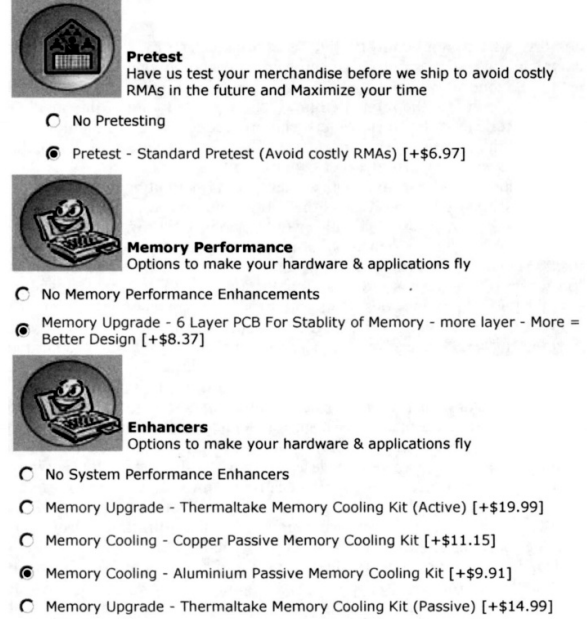


FIGURE 3.—(Continued.)

(b) Figure 3, continuation.

Figure 3: Another website designed to induce consumers to upgrade and/or buy add-ons.

7.4 Table I: Summary Statistics for Memory Module Data

Read:

- The table reports 683 website-day observations for the 128MB PC100 category.
- The lowest Pricewatch price averages \$62.98, with a standard deviation of \$33.31.
- The retailer's low-, medium-, and high-quality prices average \$66.88, \$90.71, and \$115.19.
- Average daily unit sales are much larger for low-quality modules than for medium- or high-quality modules.
- The rank variable has large variation, giving the paper power to estimate search-position effects.

Detailed interpretation: The summary statistics show the empirical ingredients of the paper. Prices are volatile, reflecting the memory market. The quality tiers are widely separated in price. Quantities are skewed toward low-quality products, consistent with the idea that Pricewatch is used to locate cheap offers. But medium and high tiers are still sold in positive quantities, making it possible to test upgrading and cross-quality demand effects.

Economic message: The table shows why the setting is powerful: there is enough price, rank, and quantity variation to study both pure price competition and obfuscation.

TABLE I
SUMMARY STATISTICS FOR MEMORY MODULE DATA (128MB PC100 MEMORY MODULES;
683 WEBSITE DAY OBSERVATIONS)

Variable	Mean	Stdev	Min	Max
LowestPrice	62.98	33.31	21.00	120.85
Range 1-12	6.76	2.52	1.00	13.53
$PLow$	66.88	34.51	21.00	123.49
$PMid$	90.71	40.10	35.49	149.49
PHi	115.19	46.37	48.50	185.50
$\log(1 + PLowRank)$	1.86	0.53	0.69	3.26
$QLow$	12.80	17.03	0	163
$QMid$	2.44	3.33	0	25
QHi	2.02	3.46	0	47

¹⁴Here, "product" also includes the quality level, for example, a high-quality 128MB PC100 module.

¹⁵Transaction prices are unavailable for products which have zero sales on a given day. These are filled in using the data collected from Pricewatch or imputed using prices on surrounding days and prices charged by the firm's other websites.

¹⁶Data are occasionally missing due to failures of the program we used to collect data and

Table I: Summary statistics for memory module data.

TABLE II
DEMAND FOR 128MB PC100 MEMORY MODULES^a

Independent Variables	Dep. Var.: Quantities of Each Quality Level		
	Low q	Mid q	High q
$\log(1 + PLowRank)$	-1.29* (10.9)	-0.77* (4.6)	-0.51* (2.9)
$\log(PLow)$	-3.03 (2.3)	-0.59 (0.4)	1.49 (0.9)
$\log(PMid)$	0.68 (0.8)	-6.74* (5.9)	2.38 (1.7)
$\log(PHi)$	0.17 (0.2)	2.72 (1.8)	-4.76* (3.3)
SiteB	-0.25* (3.5)	-0.31* (2.9)	-0.59* (5.6)
Weekend	-0.49* (8.4)	-0.94* (8.3)	-0.72* (5.8)
$\log(LowestPrice)$	1.20 (1.1)	0.83 (0.6)	-0.14 (0.1)
Number of obs.	683	683	683

^a Absolute value of t -statistic in parentheses. Asterisks (*) denote significance at the 68% level.

Table II: Demand for 128MB PC100 memory modules.

7.5 Table II: Demand for 128MB PC100 Memory Modules

Read:

- The coefficient on $\log(1 + PLowRank)$ is negative and significant for low-, medium-, and high-quality quantities.
- Moving down the Pricewatch list sharply reduces low-quality sales and also reduces higher-quality sales.
- $\log(PLow)$ has a negative and significant coefficient for low-quality demand.
- $\log(PMid)$ has a large negative significant coefficient for medium-quality demand.
- $\log(PHi)$ has a large negative significant coefficient for high-quality demand.

Detailed interpretation: Table II provides the first formal evidence that Pricewatch matters. The low-quality rank is not just a predictor of low-quality purchases; it also predicts medium- and high-quality purchases. This is the empirical signature of the loss-leader channel. Consumers use the low-quality search result to choose which site to visit, then may upgrade once on the site.

Economic message: The low advertised price is a traffic-acquisition device. It affects sales of products that are not directly advertised on the search engine.

7.6 Table III: Price Elasticities for Memory Modules

Read:

- The low-quality own-price elasticity is extremely large in magnitude: about -24.9 for 128MB PC100 and -33.1 for 128MB PC133.
- Low-quality prices also affect demand for medium- and high-quality products through the rank channel.
- Medium- and high-quality own-price elasticities are smaller in magnitude and less consistently significant.
- The patterns are present across the four memory categories.

Detailed interpretation: Table III converts coefficients into economic elasticities. The large low-quality elasticities show that search engines can create intense price competition when the product and ranking dimension are transparent. But the weaker medium and high elasticities show that the discipline is incomplete once consumers move beyond the advertised low-quality item.

Economic message: Search creates frictionless competition for the exact attribute the search engine ranks, but not for the full product bundle.

PRICE ELASTICITIES FOR MEMORY MODULES: THREE QUALITIES IN EACH OF FOUR PRODUCT CLASSES^a

	128MB PC100 Modules			128MB PC133 Modules		
	Low	Mid	Hi	Low	Mid	Hi
<i>P</i> Low	-24.9*	-12.5*	-7.2*	-33.1*	-11.2*	-4.9*
<i>P</i> Mid	0.7	-6.7*	2.4	0.8	-3.6*	0.5
<i>P</i> Hi	0.2	2.7	-4.8*	0.2	-4.8*	-4.8*
	256MB PC100 Modules			256MB PC133 Modules		
	Low	Mid	Hi	Low	Mid	Hi
<i>P</i> Low	-17.4*	-8.1*	-4.1	-24.8*	-12.5	-6.6
<i>P</i> Mid	5.7	-7.8	-4.1	0.3	3.3	3.9*
<i>P</i> Hi	0.7	6.4	-3.8	-0.9	-7.2	-0.8

^a Asterisks (*) denote significance at the 5% level.

Table III: Price elasticities for memory modules.

EVIDENCE OF INCOMPLETE CONSUMER LEARNING: CONDITIONAL SITE CHOICES OF CONSUMERS OF MEDIUM- AND HIGH-QUALITY MEMORY^a

Independent Variables	Dependent Variable: Dummy for Choice of Site A	
	Medium Quality	High Quality
$\Delta \log(1 + P_{\text{LowRank}})$	-0.64* (4.2)	-0.31* (4.0)
$\Delta \log(P_{\text{Mid}})$	-3.08* (2.2)	1.48 (1.4)
$\Delta \log(P_{\text{Hi}})$	-1.43 (1.2)	-5.73* (3.4)
Number of obs.	4118	6768

^a The table presents estimates of logit models. The dependent variable for the transaction-level data set is a dummy for whether a consumer chose to buy from Site A (versus Site B). The samples are all purchases of medium- or high-quality modules from Site A or Site B. Absolute values of *z*-statistics in parentheses. Asterisks (*) denote significance at the 5% level. The regressions also include unreported category dummies, a linear time trend, and the difference between dummies for appearing on Pricewatch's first screen.

Table IV: Evidence of incomplete consumer learning: conditional site choices of consumers of medium- and high-quality memory.

7.7 Table IV: Evidence of Incomplete Consumer Learning

Read:

- The dependent variable is a dummy for whether a consumer chose Site A rather than Site B.
- Conditional on buying medium-quality memory, the difference in low-quality rank has a significant coefficient of -0.64 .
- Conditional on buying high-quality memory, the low-quality rank coefficient remains significant at -0.31 .
- Medium- and high-quality own prices also matter, but low-quality rank continues to matter after conditioning on purchased-quality prices.

Detailed interpretation: Because the two sites sell identical product menus, this table is a strong test of consumer learning. If consumers fully compared prices for the product they bought, the low-quality rank should not matter conditional on the relevant medium or high price. Its continued effect means that the Pricewatch ranking shapes consumer attention and site choice even after consumers move away from the lowest-quality product.

Economic message: Consumers do not fully unravel obfuscation. Search rank remains a powerful attention device even for purchases of products not directly ranked.

7.8 Table V: Instrumental Variables Estimates of PC100 128MB Memory Demand

Read:

- The first three columns use cost instruments for low-, medium-, and high-quality prices.
- The second three columns use other-speed instruments, including related category prices and rank.
- The rank effect remains large and negative in several specifications.
- IV price coefficients are noisier and often imprecise, especially for cross-price effects.
- The IV results do not overturn the baseline conclusions, but they also do not provide a very sharp independent validation of all price effects.

Detailed interpretation: Table V is a robustness exercise against price endogeneity. The instruments are plausible but imperfect. The cost instruments may still reflect broader wholesale demand conditions, and other-speed instruments may share category-level shocks. The important point is that the core rank-demand relationship survives reasonably well, while precise cross-price estimates become more difficult.

Economic message: The paper’s main empirical result is not driven solely by naive price variation; the list-position mechanism remains visible when the authors use alternative instruments.

TABLE V
INSTRUMENTAL VARIABLES ESTIMATES OF PC100 128MB MEMORY DEMAND MODEL^a

Independent Variables	Dependent Variables: Quantity of 128MB PC100 Memory Modules of Each Quality Level					
	Cost Instruments			Other Speed Instruments		
	Low	Mid	Hi	Low	Mid	Hi
log(1 + PLowRank)	-1.99* (3.8)	-1.06* (2.2)	-1.19* (3.1)	-1.75* (3.0)	-0.25 (0.4)	0.02 (0.0)
log(PLow)	0.94 (0.1)	3.54 (0.6)	12.75* (2.3)	2.68 (0.6)	-1.58 (0.4)	2.39 (0.5)
log(PMid)	12.61 (1.6)	-6.88 (0.9)	0.76 (0.1)	-1.72 (1.1)	-7.14* (4.1)	-0.50 (0.1)
log(PHi)	6.41 (1.3)	6.38 (1.4)	-4.69 (1.4)	5.21* (2.5)	2.03 (0.8)	-3.45 (0.9)
SiteB	-0.40* (2.6)	-0.36* (3.0)	-0.59* (4.4)	-0.25* (2.1)	-0.51* (2.8)	-0.66* (2.6)
Weekend	-0.55* (5.1)	-0.95* (8.0)	-0.76* (4.7)	-0.48* (7.3)	-0.95* (8.2)	-0.64* (4.6)
log(LowestPrice)	-5.91 (1.0)	-2.47 (0.5)	-8.26 (1.6)	-3.89 (1.1)	2.17 (0.7)	2.37 (0.7)
Number of obs.	683	683	683	608	608	608

^aAbsolute value of *t*-statistics in parentheses. Asterisks (*) denote significance at the 5% level.

Table V: Instrumental variables estimates of PC100 128MB memory demand model.

TABLE VI
MEAN PERCENTAGE MARKUP IN SIX PRODUCT CLASSES^a

	Product Category			
	128MB Memory		256MB Memory	
	PC100	PC133	PC100	PC133
Actual low markup	-0.7%	-2.5%	4.3%	2.9%
Actual mid markup	17.3%	15.6%	16.2%	19.9%
Actual hi markup	27.3%	26.9%	24.3%	24.9%
Overall markup	7.7%	11.5%	12.7%	15.8%
Overall elasticity ϵ	-23.9	-27.7	-16.0	-21.2
1/ ϵ	4.2%	3.6%	6.3%	4.7%
Adverse selection multiplier	2.0	3.5	1.7	2.4
Predicted markup	8.3%	12.8%	10.9%	11.4%

^aThe table presents revenue-weighted mean percentage markups for products sold by websites A and B in each of four product categories along with predicted markups as described in Sections 2.2 and 7.

Table VI: Mean percentage markup in product classes.

7.9 Table VI: Mean Percentage Markup in Six Product Classes

Read:

- Low-quality markups are near zero or negative for 128MB products but positive for 256MB products.
- Medium-quality markups are around 15–20 percent.
- High-quality markups are around 24–27 percent.
- Overall markups range from 7.7 percent to 15.8 percent across categories.
- Simple inverse-elasticity predictions imply much smaller markups, but the adverse-selection multiplier raises predicted markups closer to observed values.

Detailed interpretation: Table VI ties the demand evidence to profitability. If every consumer behaved like a single-product price shopper, the low-quality elasticities would imply tiny markups. But high margins on upgrades and the adverse-selection effect mean that low-price competition is not fully passed through to the whole bundle. The adverse-selection multipliers, ranging from about 1.7 to 3.5, quantify how attracting worse-margin consumers weakens price-cutting incentives.

Economic message: Obfuscation is not merely decorative complexity. It is large enough to sustain economically meaningful markups in a price-search-dominated market.

8 Result map

- **Result 1: Price search works.** Low-quality products at the top of the Pricewatch list face extreme price sensitivity.

- **Result 2: Price search is incomplete.** Consumers do not fully use the Internet to compare all product attributes and contract terms.
- **Result 3: Loss leaders matter.** A better rank for the low-quality product increases sales of medium- and high-quality products.
- **Result 4: Add-on/adverse-selection logic matters.** A lower low-quality price changes the quality mix of buyers, reducing the profitability of aggressive cuts.
- **Result 5: Markups remain high.** Actual markups substantially exceed the inverse-elasticity benchmark.
- **Result 6: Search and obfuscation coevolve.** The search engine standardizes one dimension, and firms invest in other dimensions where consumers have difficulty comparing offers.

9 Short summary

- The paper studies Internet price search and retailer obfuscation using Pricewatch data on memory modules.
- It combines hourly price-rank data with actual sales quantities and cost data from one retailer.
- The authors document practical obfuscation strategies: complicated descriptions, quality degradation, shipping fees, upgrades, add-ons, and preselected options.
- Demand for low-quality items is extremely price sensitive; estimated own-price elasticities can exceed 20 in absolute value.
- Low-quality rank affects sales of medium- and high-quality products, showing that low-price products serve as traffic generators.
- Conditional choice tests show that consumers do not fully learn all relevant prices and terms after arriving at a site.
- IV estimates are noisier but broadly consistent with the core rank-demand findings.
- Observed markups are higher than simple inverse-elasticity rules would imply.
- Adverse-selection and add-on pricing help explain how positive markups survive even in a search-engine market.
- The paper's central lesson is that Internet search does not eliminate strategic complexity; it changes where firms compete and where they hide margins.

10 Conclusion

10.1 Core conclusion

Ellison and Ellison show that the Internet does not automatically produce frictionless commerce. Pricewatch makes low advertised prices extremely transparent and creates very elastic demand for the products ranked by the search engine. But retailers respond by designing low-quality loss leaders, upgrade menus, and add-ons that shift profits to less transparent margins.

10.2 Economic intuition

The central economic intuition is that consumers search along the dimensions made easy by the platform. Pricewatch makes the posted low-quality price easy to compare. Retailers therefore compete strongly on that dimension, but this makes it profitable to make other dimensions harder to compare. Once a consumer

arrives at a website, the retailer can use product quality differences, warranty terms, return policies, shipping, and add-ons to restore market power.

10.3 Why the evidence is persuasive

The paper’s evidence is persuasive because it links three layers of evidence. First, the screenshots show concrete obfuscation practices. Second, the demand estimates show extreme price sensitivity for low-quality products and cross-quality demand effects. Third, the markup evidence shows that obfuscation is profitable enough to matter. The result is not just that websites look confusing; confusion changes demand and markups.

10.4 Incremental contribution to the literature

The paper extends the Internet price-search literature beyond price dispersion. It directly estimates demand elasticities from transactions, connects search-theoretic and add-on-pricing theories of obfuscation, and shows that firms can endogenously degrade product quality to win search-engine rankings. This makes it an early empirical foundation for later work on shrouded attributes, drip pricing, platform design, and algorithmic ranking incentives.

10.5 Limitations

The data come from one retailer in a highly specialized market. Memory modules are technical goods sold to relatively sophisticated shoppers, and Pricewatch customers are likely more price sensitive than the average online shopper. Some price endogeneity concerns remain, and the IV estimates are imprecise. The paper also does not observe every consumer’s search path or all rival websites’ internal upgrade menus.

10.6 Takeaway

The paper’s durable message is that transparency in one dimension can create opacity in another. Platforms that rank offers by price may intensify price competition, but they may also encourage firms to design products and contracts that make true price-quality comparison harder. Search technology reduces some frictions while creating incentives for new ones.